



Vishal S R 250701844-2025-CSE ▾

V2**Started on** Friday, 7 August 2026, 9:10 AM**State** Finished**Completed on** Friday, 7 August 2026, 9:33 AM**Time taken** 23 mins 6 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void Factor( int num)
3  {
4      int count=1;
5
6      for (int i = 1; i <= num; ++i)
7      {
8          count+=2;
9          if (num % i == 0)
10         {
11             count++;
12         }
13     }
14     printf("%d", count);
15 }
16
17
18 int main(){
19     int num;
20     scanf("%d",&num);
21     Factor(num);
22 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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